

Q.1) Write a C program that illustrates suspending and resuming processes using signals

[10 Marks ]

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <signal.h>

#include <sys/wait.h>


int main() {

    pid_t pid = fork();


    if (pid < 0) {

        perror("fork");

        exit(1);

    }


    if (pid == 0) {

        // Child process: runs an infinite loop

        while (1) {

            printf("Child process is running...\n");

            sleep(1);

        }

    } else {

        // Parent process

        sleep(3);

        printf("\nParent: Suspending child process...\n");

        kill(pid, SIGSTOP); // Suspend child

        sleep(3);


        printf("Parent: Resuming child process...\n");

        kill(pid, SIGCONT); // Resume child

        sleep(3);

    }

}
```

```

    printf("Parent: Terminating child process...\n");
    kill(pid, SIGTERM); // Terminate child
    wait(NULL);
    printf("Parent: Child terminated.\n");
}

return 0;
}

```

Q.2) Write a C program to Identify the type (Directory, character device, Block device, Regular file, FIFO or pipe, symbolic link or socket) of given file using stat() system call.

[20 Marks ]

```

#include <stdio.h>
#include <stdlib.h>
#include <sys/stat.h>

int main(int argc, char *argv[]) {
    if (argc != 2) {
        printf("Usage: %s <filename>\n", argv[0]);
        return 1;
    }

    struct stat st;
    if (stat(argv[1], &st) == -1) {
        perror("stat");
        return 1;
    }

    printf("File: %s\n", argv[1]);
    printf("File type: ");

    if (S_ISREG(st.st_mode)) printf("Regular file\n");
    else if (S_ISDIR(st.st_mode)) printf("Directory\n");
    else if (S_ISCHR(st.st_mode)) printf("Character device\n");

```

```
else if (S_ISBLK(st.st_mode)) printf("Block device\n");
else if (S_ISFIFO(st.st_mode)) printf("FIFO/pipe\n");
else if (S_ISLNK(st.st_mode)) printf("Symbolic link\n");
else if (S_ISSOCK(st.st_mode)) printf("Socket\n");
else printf("Unknown\n");

return 0;
}
```